

Candidate Screening & Assessment Bot

Product Requirements Document | v0.1 | Draft

Author: ASHU KUMAR (AIML) Date: July 2026 Status: Discovery

Problem Statement

Provana's Talent Acquisition team manually screens resumes and conducts Voice & Aptitude (V&A) assessments, requiring 3 recruiters and 1 V&A resource. This creates a capacity ceiling, limits hiring scale, and costs Rs. 51 Lakhs annually. This bot automates screening and assessment end-to-end, reducing cost and nearly doubling throughput with the same headcount.

Phase 1 - Resume Screening & Bulk Upload

Scope

- Bulk resume upload: PDF, Word, Image files, Scanned PDFs
- AI-based JD matching and scoring using Claude Haiku 4.5
- Threshold logic: 10 or fewer resumes skip scoring, all go to assessment
- More than 10 resumes: LLM scores and shortlists top candidates
- Recruiter dashboard with ranked candidate results and reasons

File Handling and Edge Cases

Scenario	How We Handle It
Normal PDF	PyMuPDF extracts text directly
Scanned PDF with no text layer	Tesseract OCR reads the image layer
Image resume (JPG or PNG)	Tesseract OCR converts image to text
PDF with passport photo inside	PyMuPDF skips image blocks, extracts text only
Corrupt or unreadable file	Flagged to recruiter dashboard, skipped from processing
Non-resume file uploaded	Claude detects it is not a resume and rejects with reason

LLM Choice and Rationale

Model	Role and Reason
Claude Haiku 4.5	Primary model: resume parsing, scoring, structured extraction.

	Cheapest and fastest.
Claude Sonnet	Fallback only: used for complex or ambiguous resumes where Haiku confidence is low.

Threshold Logic

- 10 or fewer resumes uploaded => skip LLM scoring, send all to assessment directly
- More than 10 resumes => LLM scores each resume vs the JD, shortlists top X%
- Shortlist percentage (X%) to be confirmed once open questions are answered

Open Questions - Input Needed Before Architecture is Finalised

#	Question	Why It Matters
1	Standalone tool or integrated with Provana / PeopleStrong?	Core architecture decision
2	Auth: Provana SSO / Azure AD or simple custom login?	Determines user access design
3	What % of applicants receive the assessment link?	Sets the shortlist threshold
4	Notification channel: WhatsApp, email, or both?	Determines third-party integrations
5	How do candidates submit resumes: portal, email dump, or job board?	Determines ingestion pipeline

Phase 2 - Aptitude Assessment (Problem Statement Only)

Shortlisted candidates currently take manual aptitude tests administered by the team. There is no standardization and results are tracked on spreadsheets with no auto-scoring. Need a digital, auto-scored MCQ engine with randomized question banks, timed tests, and band classification: Strong, Average, or Reject.

Phase 3 - Voice Assessment (Problem Statement Only)

V&A assessment is done via manual phone calls by a dedicated resource. No scalability, subjective scoring, and no recordings or audit trail. Need an AI-driven voice test that evaluates communication clarity, fluency, and accent neutrality via speech-to-text analysis. Results should be auto-scored, recorded, and attached to the candidate profile in the dashboard.

Note: This is a v0.1 discovery document. Architecture, sprint plan, and final tech stack will be confirmed after the open questions above are resolved.